Liver Dialysis

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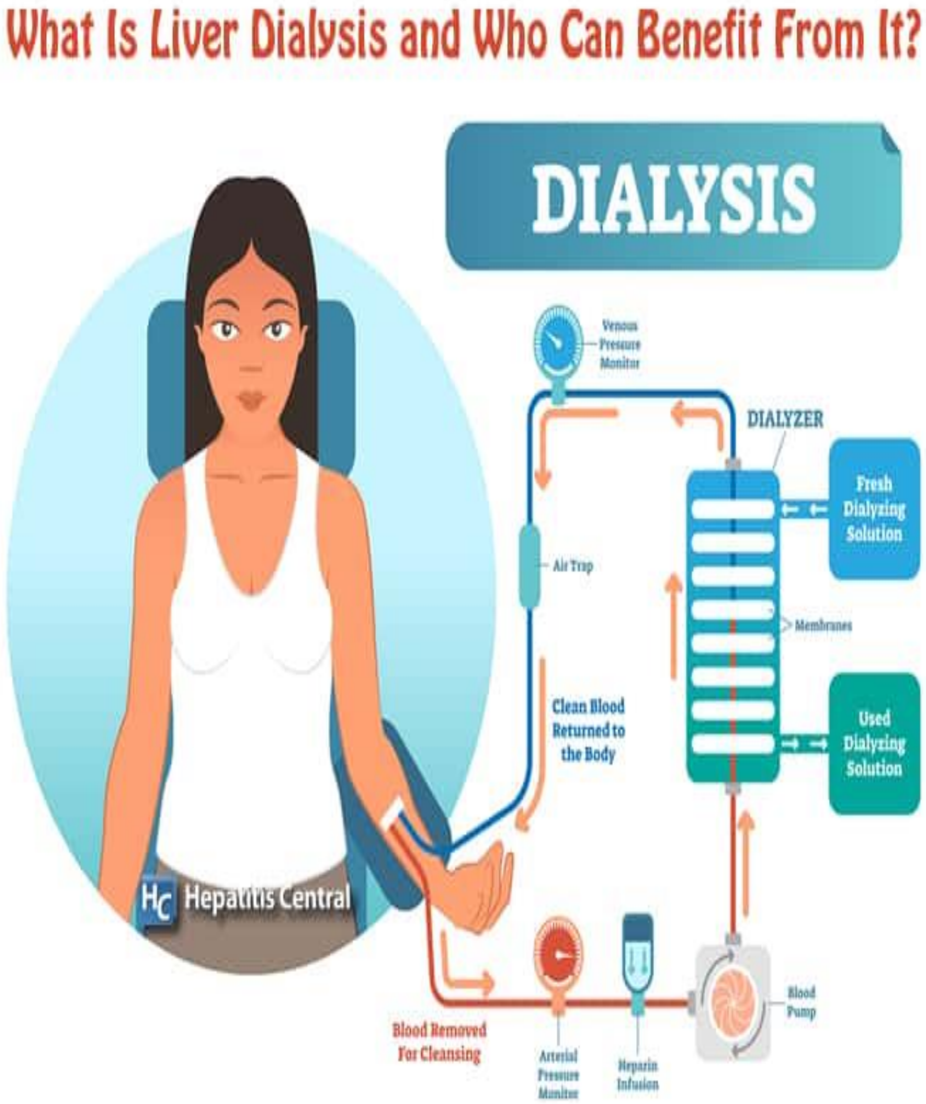
Introduction

The lack of liver detoxification as well as the loss of metabolic and regulatory functions of the liver leads to life-threatening complications, During the acute phase of their disease, a substantial number of patients are treated using extracorporeal blood purification devices. Besides regular hemodialysis, the Molecular Adsorbent Recycling System (MARS) is used with increasing frequency. In that time, liver dialysis is utilized in the appropriate cases to support the patient through his/her liver failure until transplantation can be performed.

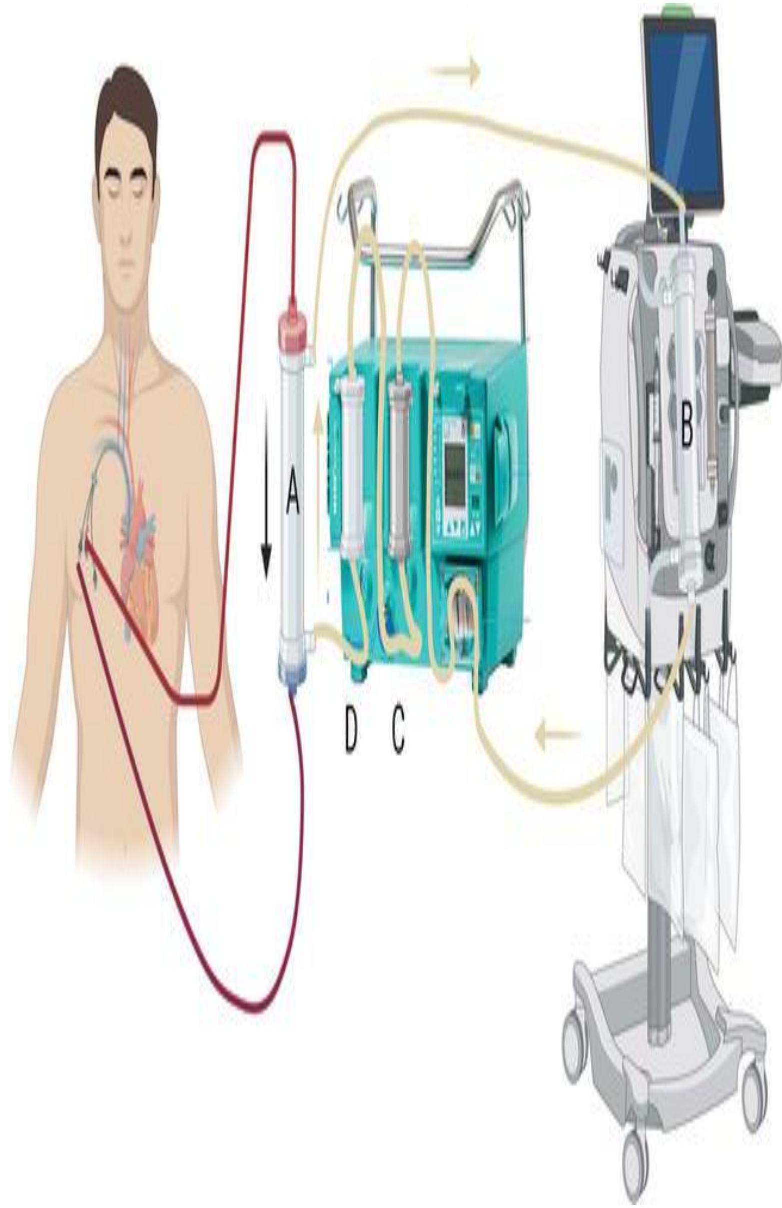
Definition of Liver dialysis

is a detoxification treatment for liver failure and has shown promise for patients with hepatorenal syndrome. It is similar to hemodialysis and based on the same principles. Like a bioartificial liver device, it is a form of artificial extracorporeal liver support.



Liver dialysis hypothesis

- **A critical issue** of the clinical syndrome in liver failure is the accumulation of toxins not cleared by the failing liver. Based on this hypothesis, the removal of
  - Lipophilic
  - Albumin-bound substances such as bilirubin, bile acids,
  - Metabolites of aromatic amino acids, medium-chain fatty acids and cytokines should be beneficial to the clinical course of a patient in liver failure.



- **Hemodialysis** is used for renal failure and primarily removes water soluble toxins, however it does not remove toxins bound

Indications for liver dialysis

- Acute hepatic encephalopathy, due to worsening of chronic liver disease (acute-on-chronic) or sudden-onset liver failure (fulminant hepatic failure) .
- Severe drug overdose, as with **acetaminophen, tricyclic antidepressants,**
- Acute decompensation of chronic liver diseases
- Acute liver failure and liver dysfunction
- Poisoning of various types including drug overdose
- Liver transplant failure
- Multiple organ failure

The goal of the liver dialysis

- Complete recovery of liver function following recompensation
- Bridging short-term or long-term waiting times until liver transplantation.
- Increasing the survival rate of liver transplants.
- Improving the quality of the patient’s life.

Dialysis Therapy can be performed in

- Transplantation centers
- Intensive care units
- In dialysis units.

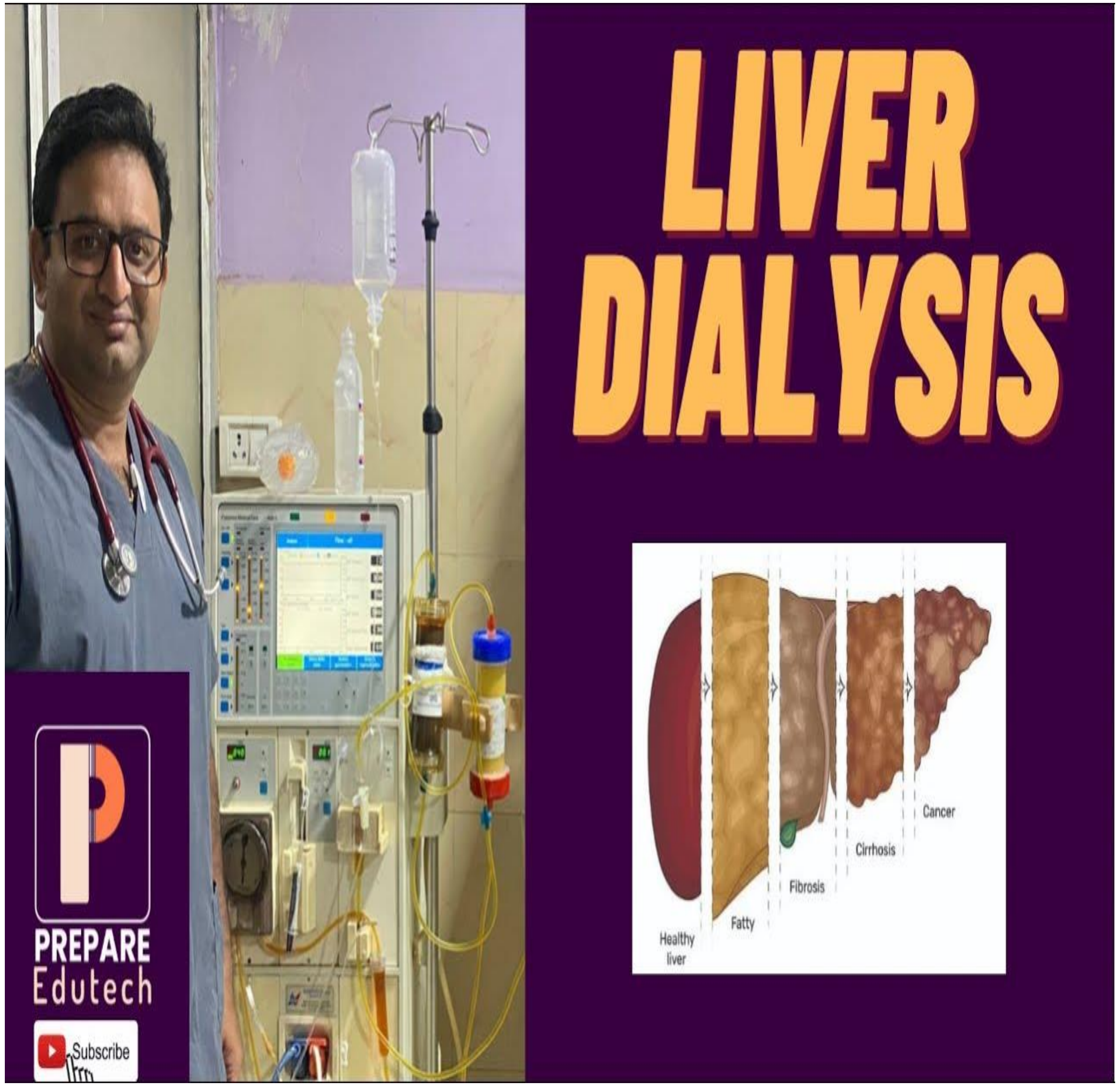
liver dialysis machines

There are broadly two categories of liver dialysis machines:

1. One that contains liver cells (biological)

2. One without any living cells (non-biological).

The non-biological version is simpler and cheaper, but it merely provides detoxifying functions without any contribution of the normal liver synthetic function.



Comparition b/n artificial and bio-artificialliver support device

Type of liver support device	Artificial liver support device	Bio-artificial liver support device
Cellular component	No	Yes
Hepatic functions achieved	Detoxification only	All hepatic functions
Cost	Comparatively less	High cost for designing, operating and managing
Ease of use	Relatively easier	Complexity of maintain living components
Efficacy	Limited	Expected results more promising

Liver dialysis devices

Artificial detoxification devices currently under clinical evaluation include the

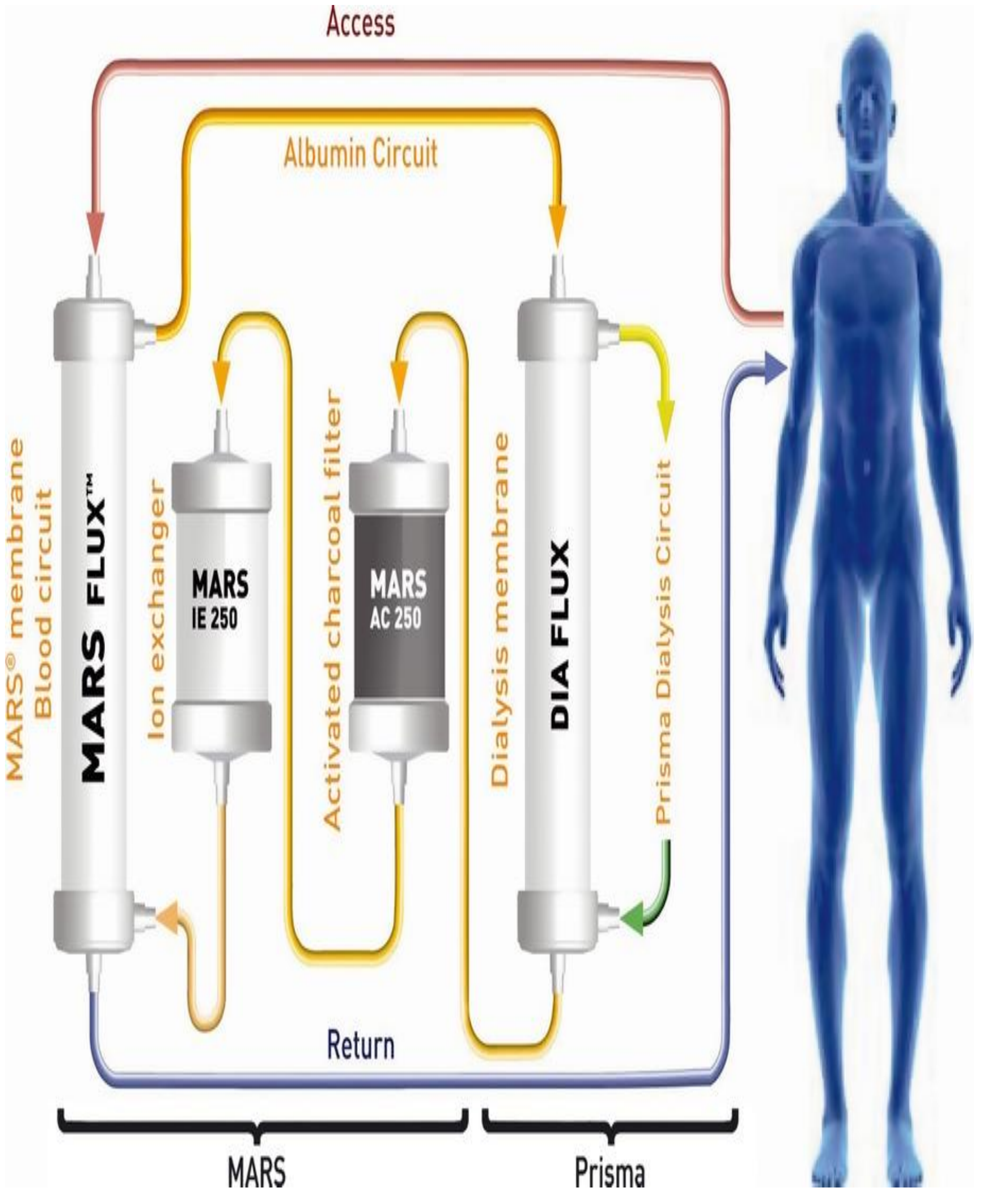
1. Molecular Adsorbent Recirculating System (MARS)
2. Single Pass Albumin Dialysis (SPAD)
3. Prometheus system.

Molecular Adsorbents Recirculation System (MARS)

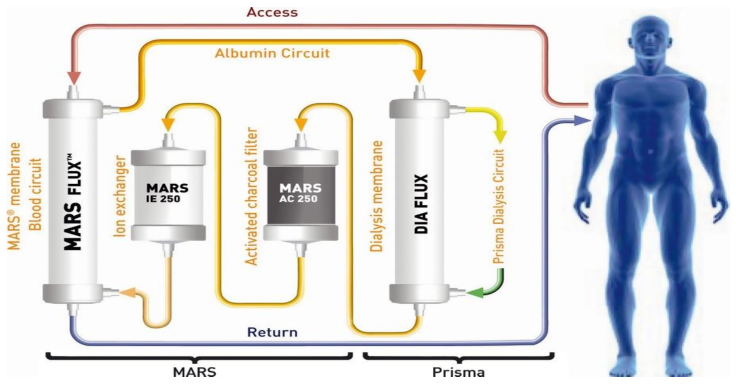
The MARS is the best known extracorporeal liver dialysis system and has existed for approximately ten years.

It consists of **two separate dialysis circuits**.

- ❑The first circuit consists of human serum albumin, is in contact with the patient's blood through a semipermeable membrane .
- ❑Two special filters to clean the albumin after it has absorbed toxins from the patient's



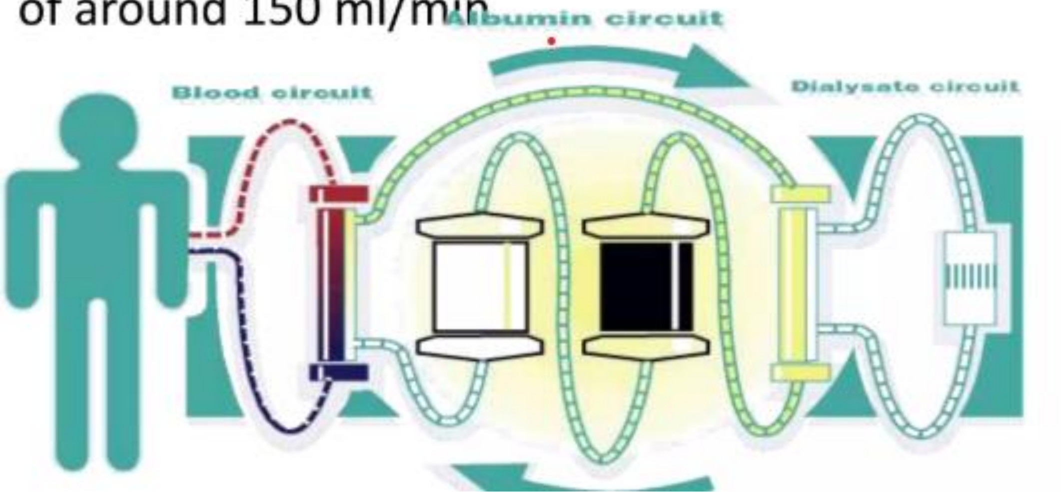
The operating principles (procedure)



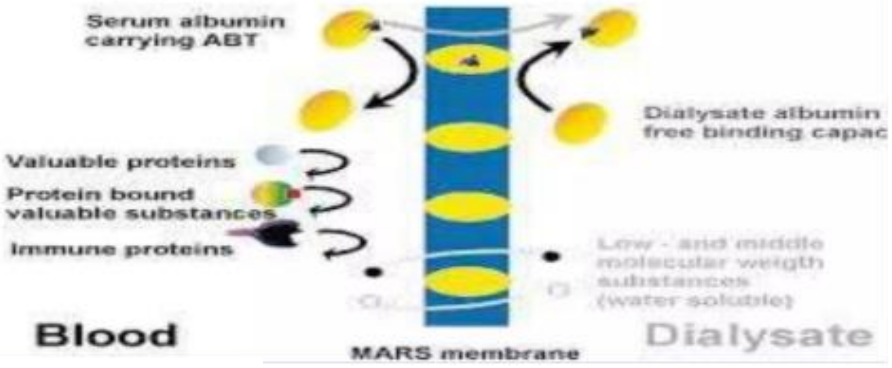
The MARS Liver Support Therapy is based on the principle of Dialysis.

- The patients blood flows through a catheter into an extracorporeal circuit equipped with a MARS flux Dialyzer made of special hollow fibres, guaranteeing a high selectivity for liver toxins.
- The blood does not get dialysed against normal dialysate but against a 20% albumin solution, albumin being a transport molecule playing an important role in human detoxification.
- The albumin dialysate will bind the toxins from the blood albumin through the membrane without exchanging any albumin molecules.
- The albumin dialysate is regenerated in a closed circuit through a second dialyser and special adsorber for repeated toxin transport.

The blood circuit generally employs a veno-venous access with a blood pump at a speed of around 150 ml/min



- Blood is passed through a special non-albumin permeable high flux dialyzer membrane usually made of polysulphone, which is capable of adsorbing albumin-bound toxins



The albumin circuit generally contains about 600 ml of 20% human albumin and is also driven by a pump at a speed of around 150 ml/mint.

This is passed through the dialysate compartment of the blood dialyzer where it removes the toxins bound to the dialyzer membrane

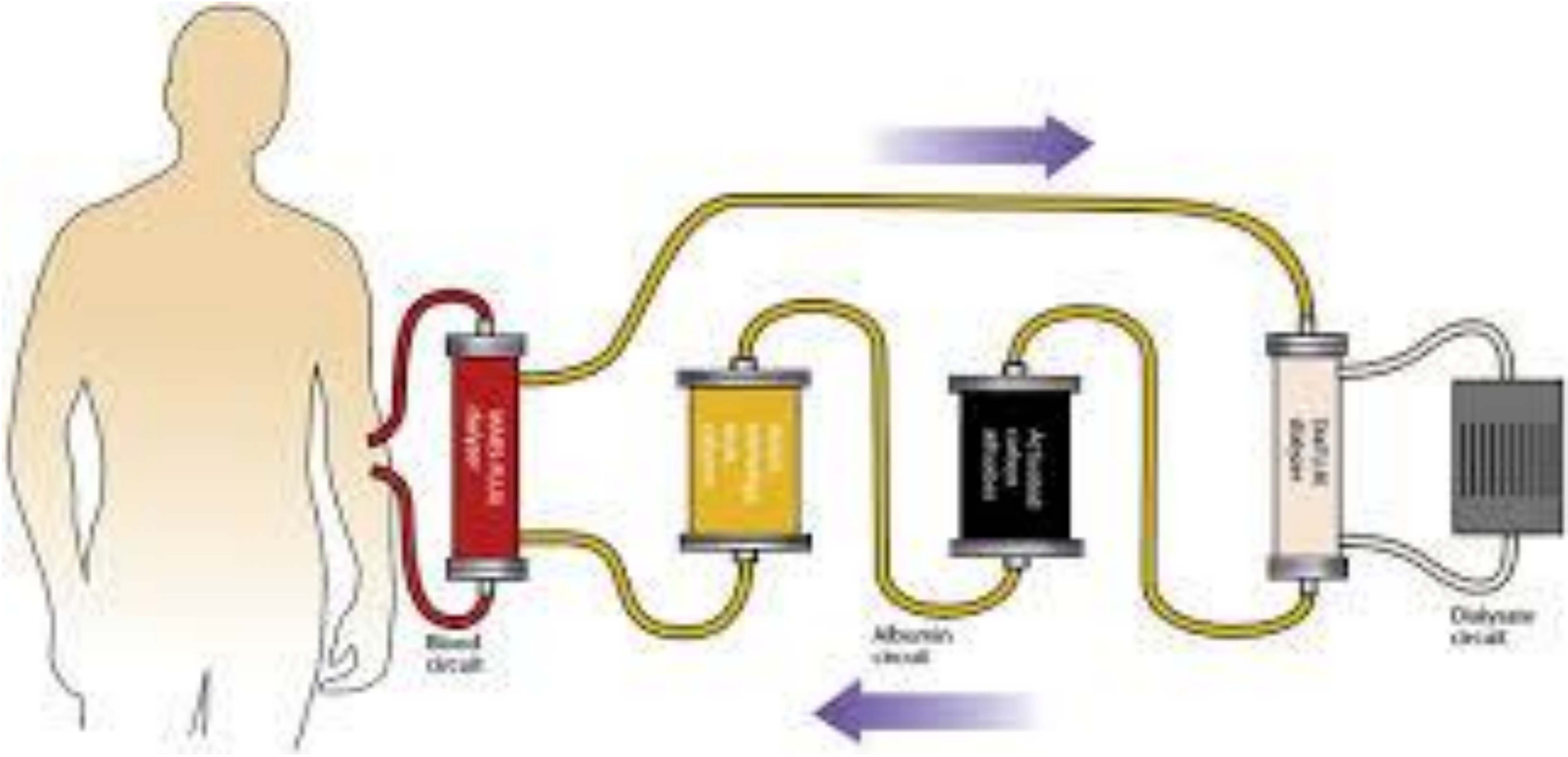
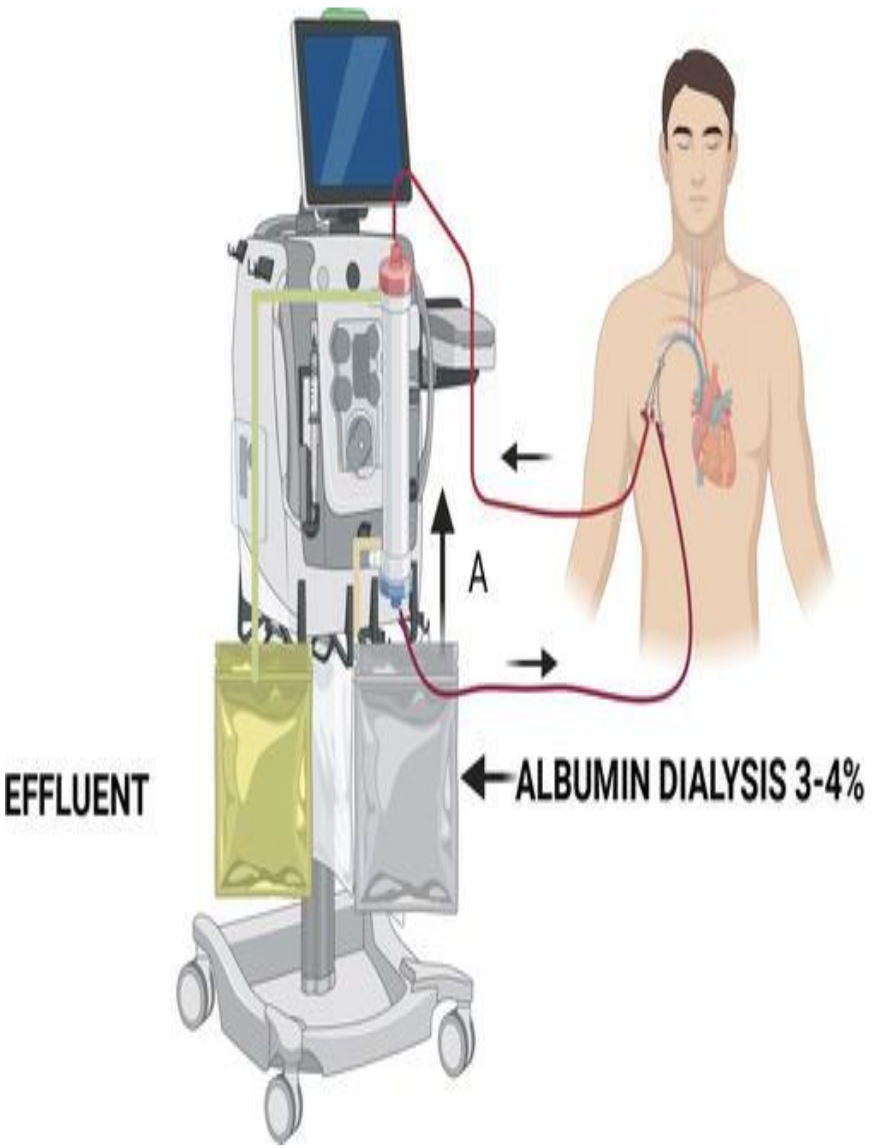


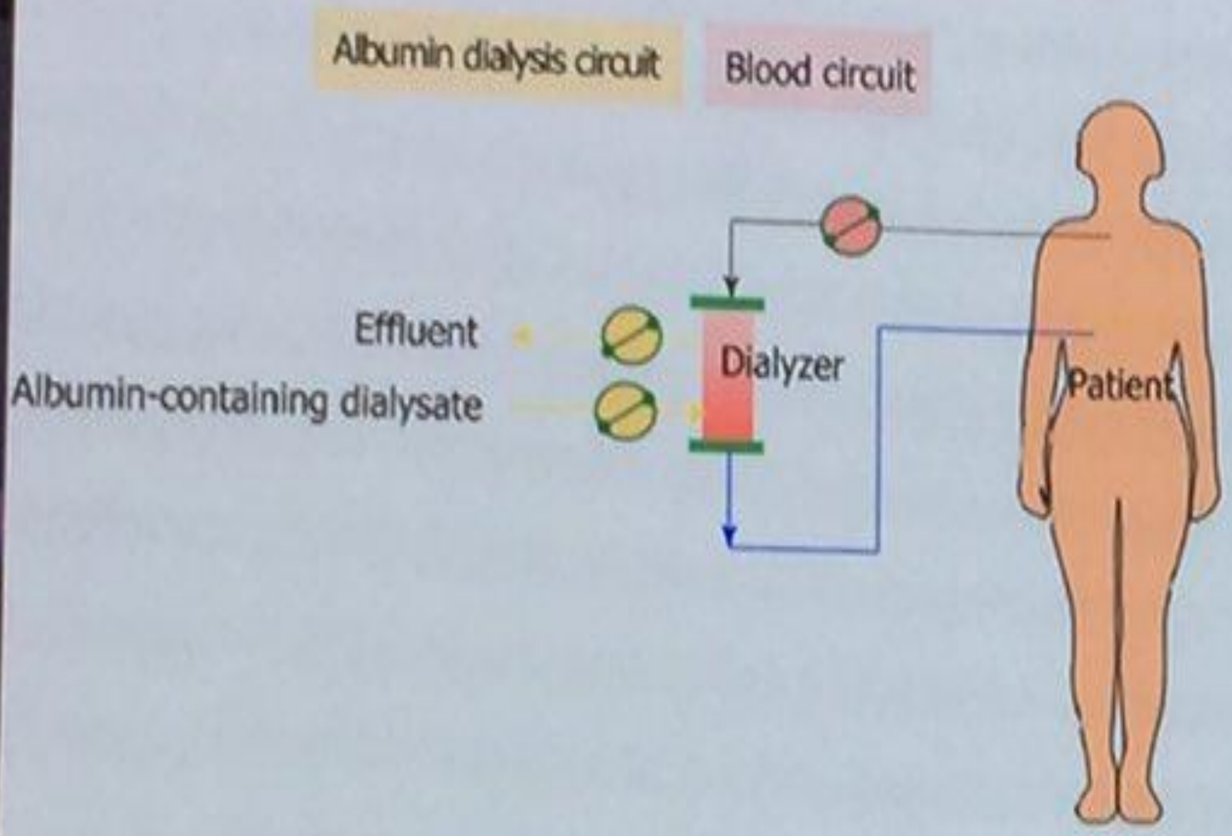
Figure 1: The MARS Circuit: Removal of Water-soluble and Albumin-bound Toxins

Single Pass Albumin Dialysis (SPAD)

SPAD is a simple method of albumin dialysis using standard renal replacement therapy machines without an additional perfusion pump system: The patient's blood flows through a circuit with a high-flux hollow fiber hemodiafilter, identical to that used in the MARS system. The other side of this membrane is cleansed with an albumin solution in counter-directional flow, which is discarded after passing the filter. Hemodialysis can be performed in the first circuit via the same high-flux hollow fibers.



SPAD –Single pass albumin dialysis



Standard CRRT circuit

3% Albumin in dialysate

Significant reduction in Bilirubin,bile acids

Prometheus

The Prometheus system is a new device based on the combination of albumin adsorption with high-flux hemodialysis after selective filtration of the albumin fraction through a specific polysulfon filter (AlbuFlow). It has been studied in a group of eleven patients with hepatorenal syndrome (acute-on-chronic liver failure and accompanying renal failure). The treatment for two consecutive days for more than four hours significantly improved serum levels of conjugated bilirubin, bile acids, ammonia, cholinesterase, creatinine, urea and blood pH. Prometheus was proven to be a safe supportive therapy for patients with liver failure.

Clinical efficiency of liver dialysis

- **Hyperbilirubinemia:** Clinically, bilirubin, which is strongly bound to albumin, is considered to be an indicator for the accumulation of albumin-bound substances in the case of

Intrahepatic cholestasias. For the first time, both bilirubin and other albumin-bound substances can be removed selectively with direct effect on hepatic cholestasis.

Clinical efficiency of liver dialysis

Hepatic Encephalopathy/Brain Edema: Ammonia is considered to be the main cause of hepatic encephalopathy (hepatic coma) and brain edema that develops as complications in liver failure. using dialysis therapy. As a result, the patients wake up, and the degree of encephalopathy is reduced.

Clinical efficiency of liver dialysis

An increased intracranial pressure causes brain stem damage; it is also one of the most frequent causes of death in cases of acute liver failure. Glutamine accumulates as a metabolic product of increased utilisation of ammonia, causing astrocytes or other cells to swell. This in turn results in brain edema. dialysis is able to stop this process and/or to reduce the intracranial pressure.

Clinical efficiency of liver dialysis

Hemodynamics: Various vasoactive substances, such as plasma nitric oxide (NO) are thought to be responsible for the characteristic hemodynamic situation in cases of liver failure. dialysis removes NO from the blood and reduces its production. This is considered to be the cause of significant increases in systemic vascular resistance (SVRI) and mean arterial pressure (MAP) following dialysis. At the same time, the heart returns to normal with a decrease in cardiac index and the local supply of blood to the liver, brain and kidneys is improved; with a reduction in pressure in the portal vein.

Clinical efficiency of liver dialysis

Renal Function: If the liver failure is accompanied by a deteriorating renal function, the prognosis is poor, with the failure of an additional organ system being a further step towards the development of multiple organ failure. Various studies have shown that this development can be reversed and even avoided using dialysis. This may be attributed to an improvement in the systemic and the local hemodynamic situation, but also perhaps to a direct effect of dialysis Therapy.

Clinical efficiency of liver dialysis

Protein Synthesis: Both the detoxification and the synthesis functions of the liver begin to improve on removal of albumin bound toxins in the presence of an adequate regenerative capacity in the damaged liver.

There is positive effect on patient neurologic status and physiologic status. Clinical experience suggests that use of the Liver Dialysis Unit early in the disease process may avert respirator use, avoid or decrease days in the ICU, shorten length of stay, and obviate or postpone the need for transplant or improve the likelihood of successful transplant by stabilizing the patient’s health prior to surgery.

During treatment, a small amount of blood is drawn from the patient and passed across the dialyser membrane in the Liver Dialysis Unit. On the other side of the membrane, the sorbent suspension serves as a "sink" that draws toxins from the blood. The treated blood is then returned to the patient.

Treatment is usually done for 2-5 consecutive days lasting 4-6 hours.

Side effects of liver dialysis

There are some long-term side effects of liver dialysis that patients should be aware of before treatment. Infection is one major side effect that can occur with any type of dialysis. Patients should be closely monitored to ensure that any infections that do occur are kept under control.

Other side effects may include weight gain, low blood pressure, iron deficiency, hernia, low blood pressure, and nerve damage. Liver dialysis patients may also experience symptoms like severe depression and anxiety, oftentimes as a result of the stress from dealing with a severe illness. Medications may be used to treat these side effects if they become a serious danger to the patient's health. Therapy may also be provided for those with psychological side effects.

Thank You